

ANALYTICAL REPORT | JUMBOLINE OPTICAL NETWORKS

CUSTOMER CHURN ANALYSIS

Data Cleaning, Exploratory Analysis, Business Insights & Retention Strategy

Portfolio analytics project | Python • PostgreSQL • Power BI

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LINDA NYAKASI
lindanyakasi@gmail.com

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Executive Summary

This analysis examines 7,043 Jumboline's customers to understand churn, identify high-risk customer configurations, quantify recurring revenue exposure, and translate the findings into retention priorities. The workflow follows a CRISP-DM-aligned progression from data understanding and cleaning through exploratory analysis, segmentation, financial interpretation, and interactive reporting.

Overall churn is 26.54% (1,869 customers). The strongest observed relationships are concentrated around contract type, internet service, payment method, and tenure. Month-to-month customers churn at 42.71%, fiber-optic customers at 41.89%, and electronic-check customers at 45.29%. The highest observed intersection is month-to-month + fiber optic at 54.61%; adding electronic check raises the observed churn rate to 60.4%.

Tenure provides the clearest lifecycle signal: churned customers have a median tenure of about 10 months versus about 38 months for retained customers, and first-year customers show substantially higher churn (47.44%) than longer-tenured groups. This makes the first 6–12 months the most important retention window in the current data.

Household structure and monthly charges provide secondary, supporting evidence. Customers without a partner or dependents show higher churn, while churned customers carry meaningfully higher monthly charges, with a median of about \$79.65 versus \$64.43 among retained customers. Gender and phone service subscription show no meaningful relationship with churn and therefore provide limited value for retention targeting.

The financial impact is also material. Customers who have churned were associated with approximately \$139,131 of monthly recurring revenue, equivalent to 30.5% of total monthly revenue in the dataset. This is revenue already lost in the observed snapshot, not a forecast of future loss.

Management Takeaway

The retention problem is not simply "high churn." It is concentrated early in the customer lifecycle and amplified by flexible contracts, fiber service, and electronic-check payment. Retention investment should therefore be targeted rather than spread uniformly across the base.

Objective and Scope

Objective

To Identify customer and service characteristics associated with churn, validate and prepare the data for downstream analytics, and convert the findings into a stakeholder-ready retention framework.

Specific Objective

To translate customer-level churn data into a defensible business story:

1. Where churn concentrates,
2. Which customer configurations deserve attention,
3. How much recurring revenue is exposed, and
4. How a retention program could be operationalized.

Scope

- ❖ Profile the raw dataset and establish its structure and grain.
- ❖ Standardize names, data types, whitespace, categorical values, and numeric fields.
- ❖ Identify missing values, duplicates, and anomalous values.
- ❖ Validate the cleaned dataset before analysis.
- ❖ Explore churn across demographic, service, contract, payment, tenure, and financial variables.
- ❖ Assess numerical relationships and potential feature redundancy.
- ❖ Combine dimensions to identify high-risk segments.
- ❖ Quantify recurring revenue exposure.
- ❖ Deliver a Power BI reporting layer and actionable retention recommendations.

Data Understanding and Preparation

The analytical table contains 7,043 customer records. customer_id is treated as an identifier, while churn is the target outcome. The core analytical fields cover household characteristics, service adoption, contract type, payment method, tenure, monthly charges, and total charges.

Data Quality Issues and Treatment

Quality Area	Issue Identified	Treatment Applied	Why it Matters
Column names	Inconsistent naming conventions across fields	Standardized to snake_case	Ensures database compatibility and prevents syntax errors during SQL querying and DAX modeling.
Whitespace	Leading and trailing spaces in text fields	Stripped from all text values	Prevents false duplicate categories and lookup join failures caused by hidden spaces.
Data types	total_charges stored as text instead of numeric	Converted to numeric after handling blank values	Enables mathematical aggregation and prevents type conversion errors in DAX measures.
Binary fields	Inconsistent coding formats across boolean columns	Converted 1/0 values in senior_citizen to Yes/No	Standardizes all binary dimensions to a unified format across report slicers, DAX logic, and filter criteria.
Categorical values	Non-standard string formatting in contract types	Standardized 'Month-to-month' to 'Month to month'	Fixes string mismatches to ensure clean categorical aggregation and uniform labeling across visuals.
Missing values	Required explicit assessment after type conversion	Checked and confirmed post cleaning	Prevents broken calculations, blank cards, or distorted averages across report visuals.
Duplicates	Required explicit assessment	Checked using the customer identifier; none removed	Protects one-row-per-customer grain, preventing inflated KPI counts and distorted churn rates.
Outliers	Potential skew in numeric metrics (tenure, charges)	Inspected distributions using summary stats; retained valid extreme values	Ensures extreme values represent legitimate high-value customers rather than data entry errors, preserving accurate revenue totals.

Post-Cleaning Validation

A final audit rechecked row count, column inventory, data types, missing values, duplicates, categorical consistency, numeric ranges, and overall shape. The validated DataFrame was treated as the analytical source for the downstream analysis rather than mixing raw and cleaned versions.

Exploratory Data Analysis

EDA is organized as a funnel: establish the baseline, understand the distributions, compare churn across individual features, examine numeric differences, then move to combined segments where the business signal becomes stronger.

Churn Baseline

Of the 7,043 customers, 5,174 are retained and 1,869 have churned. The resulting 26.54% churn rate is the benchmark against which segment-level rates should be interpreted.

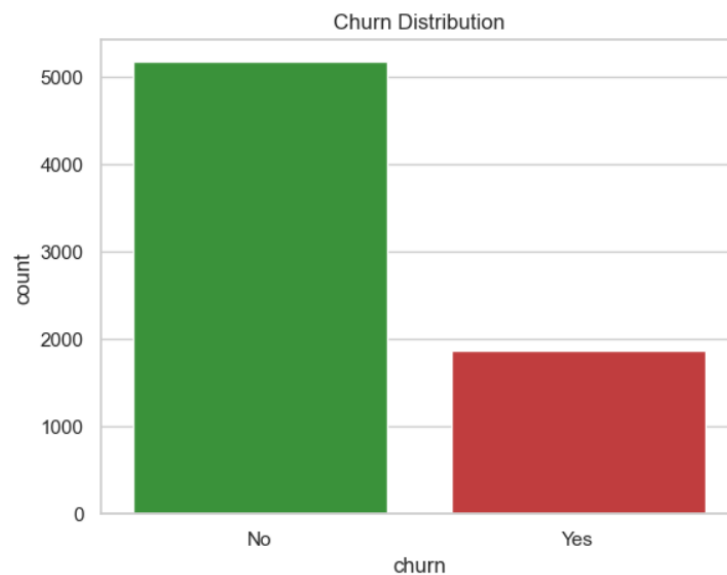


Figure 1. Overall churn distribution (No = 5,174, Yes = 1,869).

Categorical Features: Risk vs Scale

Segment prioritization should consider both churn rate and segment size. A small segment can have extreme churn without representing a large business opportunity, while a large segment with moderately elevated churn can have greater aggregate impact.



Figure 2. Churn rate by category, annotated with each segment's share of the total customer base.

- High-risk, high-volume segments
 - Month-to-month: 42.71% churn and approximately 55% of the customer base.
 - Fiber optic: 41.89% churn and approximately 44% of the customer base.
 - Electronic check: approximately 45.3% churn and approximately one-third of the customer base.
- High individual risk, lower overall impact
 - Senior citizens: approximately 42% churn but a smaller base share, only 16.2%. This segment carries high individual risk but lower scale; hence, smaller total revenue impact.
- Moderate, protective effect
 - Shared households appear to create friction against switching providers. Customers without a partner or without dependents, on the other hand, churn at noticeably higher rates than customers with a partner or dependents.
- No meaningful effect
 - Gender and phone service: little observed differentiation; these are not priority targeting variables.

Numeric Patterns

Tenure is concentrated at the low and high ends, while monthly charges show a strong low-price cluster and a broader higher-charge cluster. Total charges are right-skewed because accumulated charges naturally grow with tenure. A log transformation was tested but did not materially improve the business interpretation, so the raw value was retained for reporting.

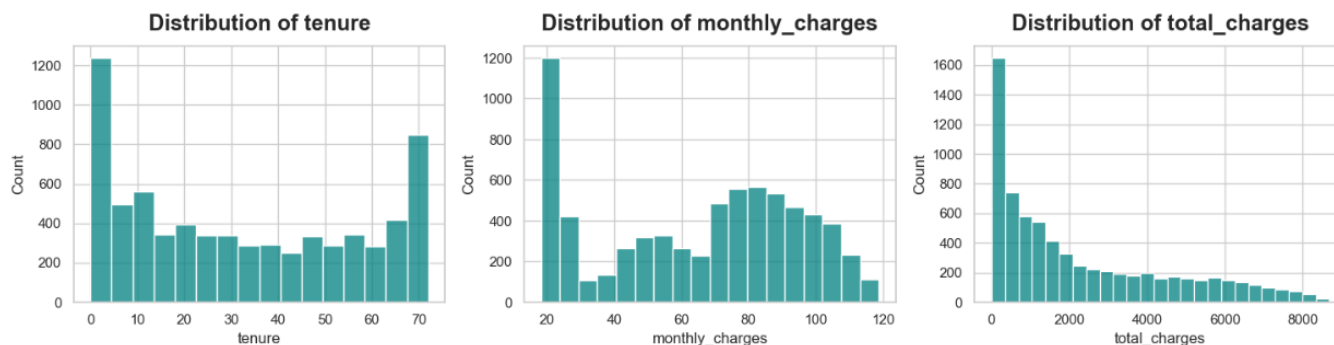


Figure 3. Distribution of tenure, monthly charges, and total charges.

Comparing churned and retained customers reinforces the lifecycle pattern: churners have much shorter tenure and higher median monthly charges. Total charges are lower among churners largely because early departure limits the time available to accumulate charges.

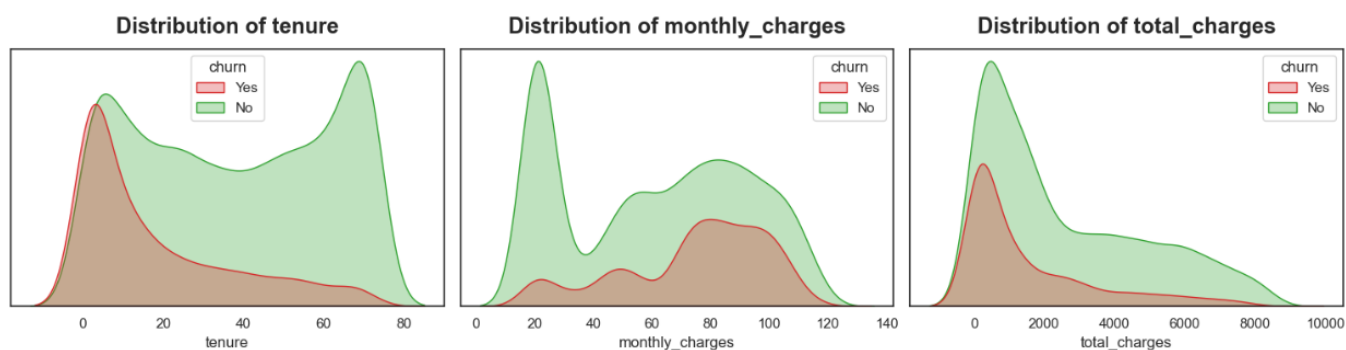


Figure 4. Density of tenure, monthly charges, and total charges, split by churn status.

Boxplots comparing churned and retained customers across tenure, monthly charges, and total charges confirm the same three-way relationship seen in the histograms and density plots, now with precise reference points.

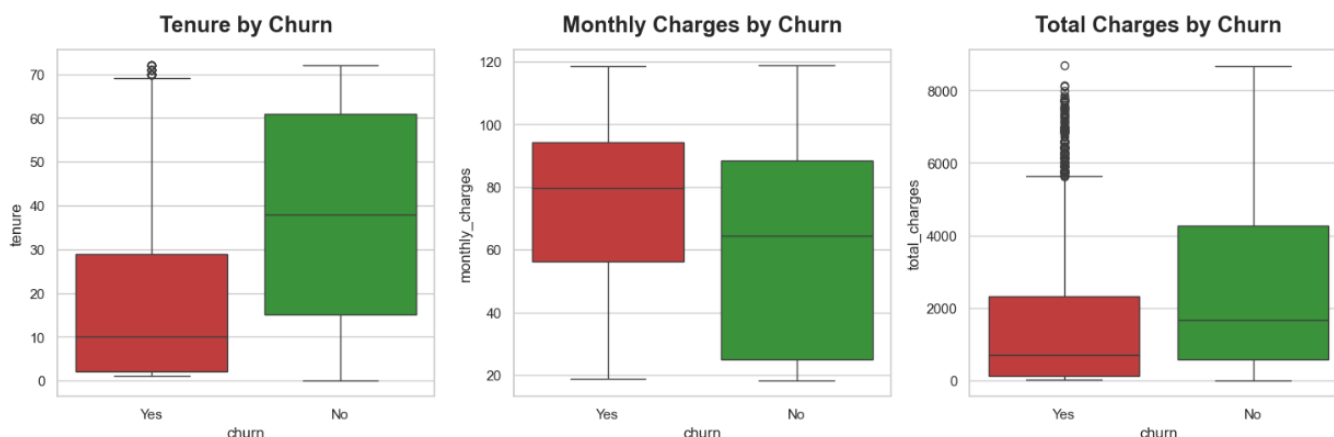


Figure 5. Tenure, monthly charges, and total charges, compared for churned versus retained customers.

Metric	Churned Customers (median)	Retained Customers (median)
Tenure	~10 months	~38 months
Monthly charges	~\$79.65	~\$64.43

Metric	Churned Customers (median)	Retained Customers (median)
Total charges	~\$703.55	~\$1,700

Interpreted together, these three metrics tell a consistent story: customers are churning early in their lifecycle, at a point where they are paying comparatively high monthly rates and have not yet accumulated meaningful lifetime value.

Tenure: The Lifecycle Signal

Grouping tenure into bands shows that churn declines sharply as tenure increases.

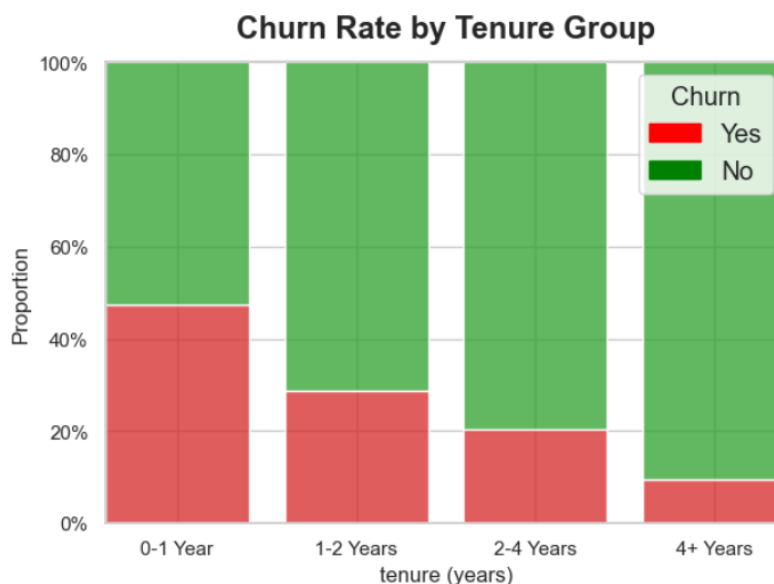


Figure 6. Churn rate by tenure group.

This is the strongest reason to treat the first 0–12 months as a distinct retention window. The pattern should be interpreted as an association in a cross-sectional dataset, not as a longitudinal survival curve.

Relationship and Redundancy Analysis

Correlation is used to understand linear association and potential redundancy, not to establish causation.

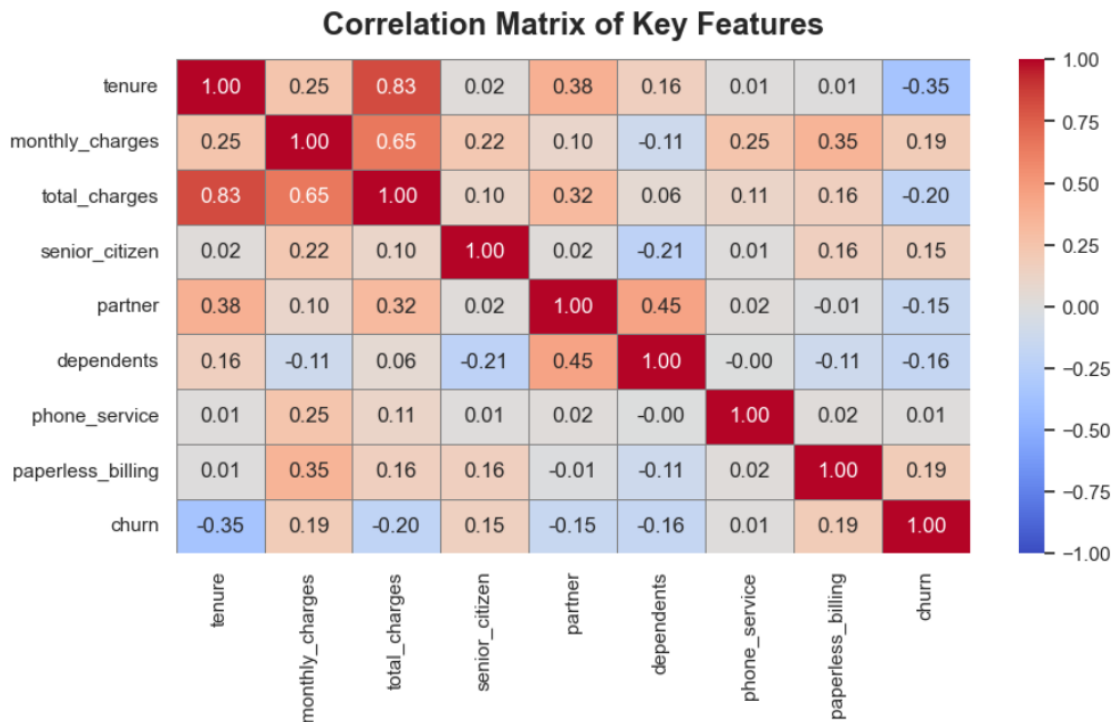


Figure 7. Correlation matrix of key numeric and binary features, including churn.

Correlation with Churn

Tenure has the strongest linear association with churn among the numeric and binary variables examined. Other numeric variables show weaker associations and should be treated as supporting signals rather than standalone drivers.

Tenure and Total Charges

Tenure and total_charges are strongly correlated at approximately +0.83. This is expected because accumulated charges depend heavily on how long a customer has been billed. Both variables are retained for descriptive reporting because they answer different business questions, but a future predictive model should assess whether both are necessary and should test for redundancy and leakage before deployment.

Advanced Segmentation: Where Risk Concentrates

Contract × Internet Service

Some variables are far more informative when combined than when read individually.

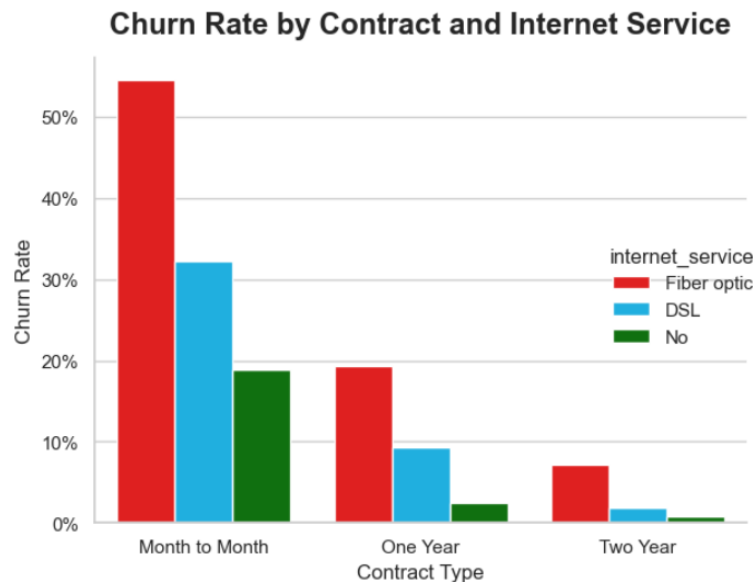


Figure 8. Churn rate by contract type and internet service.

The combination of contract and internet service reveals substantially more concentration than either variable alone. Month-to-month fiber customers churn at 54.61%, compared with 32.22% for month-to-month DSL and 18.89% for month-to-month customers with no internet service.

Fiber remains the highest-churn internet tier across contract lengths. This persistence makes the fiber customer experience a specific investigation area rather than simply a contract-duration issue.

Contract × Internet Service × Payment Method

Adding payment method sharpens the segmentation further. The month-to-month + fiber optic + electronic check segment reaches an observed churn rate of 60.4%. This is a useful targeting profile because it identifies a specific intersection rather than treating all fiber or all month-to-month customers as equivalent.

Adding a fourth factor, tenure of ≤12 months, sharpens this further. Among the 631 customers who have ever matched all four conditions (month-to-month, fiber optic, electronic check, and tenure of 12 months or less at the time of measurement), 449 have churned, a historical churn rate of 71.16%. The 182 who remain active are precisely the customers the Customer Risk Scoring framework below labels Critical. This four-factor cut is the same population described two ways, a historical rate looking backward, and a live exposure count looking forward at who is left.

Interpretation Guardrail

A high-risk intersection identifies where churn is concentrated; it does not prove that changing one attribute will cause the customer to remain.

Customer Risk Scoring

For operational reporting, the project uses a four-factor risk framework on currently active customers: month-to-month contract, fiber optic internet, electronic-check payment, and tenure of 12 months or less. One point is assigned per condition, producing a 0–4 score and corresponding Minimal, Low, Medium, High, and Critical labels. This score is a prioritization device based on observed associations, not a calibrated probability model.

Risk Tier	Definition	Active Customers
Critical	All 4 risk factors present (Risk Score = 4)	182
Retention Targets	3 or more risk factors present (Risk Score $\geq$ 3)	782

These 782 Retention Target customers account for 19.18% of the company's active monthly recurring revenue, meaning roughly one fifth of ongoing revenue sits with customers who already show three or more of the four warning signs identified throughout this report, before any of them have actually churned. This is the number the dashboard's Action List page is built to act on directly.

Key Business Findings

- **Lifecycle:** Churn is concentrated among shorter-tenure customers; the median tenure is about 10 months for churners versus 38 months for retained customers.
- **Contract:** Month-to-month customers churn at 42.71%, roughly four times the one-year rate and far above the two-year rate.
- **Internet service:** Fiber optic churn is 41.89%, more than twice the DSL rate, and remains elevated across contract lengths.
- **Payment:** Electronic check has the highest payment-method churn rate at approximately 45.3%.
- **Price signal:** Churned customers have a higher median monthly charge than retained customers, indicating a useful value/risk context signal.
- **Intersection:** Month-to-month + fiber reaches 54.61%; adding electronic check produces the highest observed three-way segment at 60.4%.
- **Business impact:** Segment size matters: retention resources should prioritize high-risk segments that also carry meaningful customer volume and recurring revenue.
- **Low-value variables:** Gender and phone service show little differentiation and should not receive disproportionate targeting attention.

Business Impact and Financial Analysis

Revenue at Risk

Churned customers account for approximately \$139,131 in monthly recurring revenue, equal to 30.5% of the dataset's total monthly revenue. This is an observed snapshot of revenue associated with customers who have already churned, not a forecast of future revenue loss.

Customer Value by Contract

Churned two-year customers are few (48 customers) but have much higher accumulated charges per customer (roughly \$5,432) than month-to-month churners (1,655 customers but average roughly \$1,164 each). This creates two different retention economics: month-to-month churn is primarily a volume problem, while long-contract churn is rarer but potentially more valuable per individual customer.

Add-On Services and Payment Method

Online Security and Tech Support are associated with substantially lower churn than several other service configurations.

Service	Churn Rate Without	Churn Rate With	Reduction
Online Security	31.33%	14.61%	16.72 points
Tech Support	31.19%	15.17%	16.02 points
Online Backup	29.17%	21.53%	7.64 points
Device Protection	28.65%	22.50%	6.15 points

Automated payment methods also show lower churn than electronic check, and are associated with meaningfully longer customer relationships. Electronic check customers churn at the highest rate of any payment method (45.3%) and also show the shortest average billing history (approximately 25 months, versus 43 to 44 months for the two automated payment methods).

Segmenting customers by whether they have a protective add on (Online Security or Tech Support), an automated payment method, both, or neither, shows a clear stacking effect:

Lever Combination	Churn Rate
Neither Add-on nor Automated payment	40.38%
Add on only	23.53%
Automated payment only	20.77%
Both Add-on and Automated payment	11.69%

Customers with both a protective add-on and automated payment show 11.69% churn versus 40.38% among customers with neither. These are observed associations and should be validated through controlled retention testing before being treated as intervention effects.

Retention Scenario

In general, a 10-percentage point reduction in churn would preserve an estimated \$13,913 in monthly recurring revenue.

Within month-to-month fiber optic customers, a hypothetical 10-percentage-point reduction in churn corresponds to approximately \$10,048 of monthly recurring revenue preserved in the current dataset; a 20-point reduction corresponds to approximately \$20,096.

Power BI Reporting Architecture

The findings in this report are also delivered as a live, interactive Power BI dashboard, connected directly to the PostgreSQL views, so it reflects the current data on every refresh. The dashboard is organized into six pages. Each page answers one business question, and each visual has a single analytical job:

1. Executive Overview — Where is the core retention risk?

Headline KPIs establish revenue exposure, total customer scale, and baseline churn. Operational hotspots sit alongside macro tenure trends to provide C-suite leadership with a quick, strategic snapshot of active business risk.

2. Churn Lifecycle — When are we losing customers?

Tracks churn concentration across granular onboarding windows from initial sign-up through multi-year accounts. Pinpoints early-stage drop-off rates in the critical 90-day onboarding zone to align lifecycle stages with financial exposure.

3. Risk Explorer — Which customer segments are most vulnerable?

Cross-analyzes churn rates across contract types, internet technologies, and tenure groups via a central Risk Matrix. Features a dynamic segment driver tool to pivot churn rates across behavioral and demographic variables on demand.

4. Revenue Impact & ROI — Where is the financial exposure and what is fixing it worth?

Maps churn rates against customer lifetime value (CLV) and total revenue loss across contract lengths. An interactive retention simulation slider calculates direct dollar savings for targeted churn reduction targets.

5. Products & Payments — How do add-on services and payment channels drive retention?

Evaluates baseline churn differences between standalone add-ons, payment channels, and bundled service suites. A decomposition tree maps sequential product journeys, proving how auto-pay paired with tech support cuts churn to its lowest point.

6. Action List — Who should we intervene on first?

Categorizes active accounts across a four-factor risk distribution from Critical to Minimal risk. Pairs risk metrics directly with a ranked customer outreach queue filtered to Critical risk

customers and sorted by average monthly charge, equipping retention teams with target customer IDs for immediate intervention.

Recommendations

- 1. Prioritize the first 6–12 months** — Create structured onboarding checkpoints, early service-health checks, and proactive outreach before the highest-risk window passes.
- 2. Reduce month-to-month exposure** — Test targeted contract-conversion offers to encourage migration of eligible month to month customers to one-year contracts rather than applying blanket discounts. Measure post-conversion churn against a control group, the migrated customers.
- 3. Investigate the fiber experience** — Join churn data with network performance, outage, installation, ticket, and complaint data to determine whether the observed gap is operational, pricing-related, or both.
- 4. Address the electronic-check journey** — Investigate why electronic-check customers churn more and test friction-reduction or payment-migration interventions. Offer a small incentive, such as a one-time bill credit, to migrate electronic check customers to automatic bank transfer or credit card payment
- 5. Target high-risk, high-value customers** — Prioritize active customers who combine multiple observed risk factors and meaningful monthly charges; use the risk score as an outreach queue, not a probability forecast.
- 6. Test protective service/payment combinations** — Online Security or Tech Support plus automated payment is a promising observed association. Validate the combination experimentally before scaling incentives.

Limitations and Next Analytical Steps

Limitations

1. The dataset is cross-sectional; it does not observe each customer continuously from acquisition through churn. Several of the recommendations above should be validated with additional operational data or controlled experiments before large scale rollout.
2. The analysis identifies associations, not causal effects.
3. Fiber service quality, outage history, support interactions, acquisition channel, and competitor context are not included.
4. The risk score is a rule-based prioritization framework, not a calibrated churn probability.
5. Retention scenario values are sensitivity calculations based on assumed churn reduction.

Next Steps

1. Investigate fiber optic service quality directly using network performance, outage, and support ticket data.
2. Add acquisition channel data to test whether certain sign-up sources produce higher risk customers.
3. Examine contract start dates and billing changes to understand exactly when churn risk peaks within a customer's history.
4. Tenure and total charges are strongly related. Apply statistical testing or predictive modeling to confirm which associations remain significant once other variables are controlled for.
5. Track retention interventions over time and measure their actual, realized effect on churn rather than assuming a fix worked.

Data Delivery and Technical Stack

Python (Pandas/NumPy) was used for cleaning, validation and exploratory analysis. SQLAlchemy was used to move the validated dataset into PostgreSQL. PostgreSQL provides persistent analytical storage and a reporting layer, while Power BI provides the interactive stakeholder-facing layer.

Technology	Role
Python / Pandas / NumPy	Cleaning, validation, transformations, and EDA
Matplotlib / Seaborn	Exploratory visual analysis
SQLAlchemy	Validated data transfer to PostgreSQL
PostgreSQL / SQL	Persistent analytical storage and business logic
Power BI / DAX	Interactive reporting, segmentation, risk scoring and scenario analysis

Conclusion

The analysis points to a concentrated retention problem rather than a uniform churn problem. Risk is highest early in the customer lifecycle and is amplified within month-to-month, fiber-optic and electronic-check segments. The business impact is material, with approximately \$139,131 of monthly recurring revenue associated with customers who have already churned.

The recommended response is targeted: strengthen the first-year customer journey, reduce month-to-month exposure, investigate the fiber experience, address payment friction, and prioritize active customers who combine multiple risk signals with meaningful recurring value. The next analytical step is not simply more visualization; it is linking churn to operational data and testing retention interventions so observed associations can be converted into measured business impact.

Appendices

Appendix A. Power BI Implementation Blueprint

Foundational measures

- Total Customers = `DISTINCTCOUNT(vw_customer_analytics[customer_id])`
- Churned Customers = `CALCULATE([Total Customers], churn = "Yes")`
- Churn Rate = `DIVIDE([Churned Customers], [Total Customers])`
- Monthly Revenue = `SUM(monthly_charges)`
- Churned Monthly Revenue = `CALCULATE([Monthly Revenue], churn = "Yes")`

Tenure segmentation

Use `tenure_group_detailed` only where the broader 0–1, 1–2, 2–4, 4+ year segmentation is more appropriate.

Use `tenure_group_rollup` only when a finer grain 0-3, 4-6, 7-12, 13-24, 24+ month segmentation is needed

Appendix B. Analytical Language Standards

Use	Avoid
associated with	caused by
observed churn rate	probability of churn
highest observed-risk segment	customers who will churn
scenario estimate	forecast
revenue associated with churned customers	future revenue loss
protective association	guaranteed retention lever
targeting hypothesis	proven intervention